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When there is a change of state from solid to liquid (melting), liquid to vapour (evaporation) or solid to vapour (sublimation) there is a change in entropy. This change may be carried out at constant temperature reversibly as two phases are in equilibrium during the change.

$$\Delta S = \frac{q_{rev}}{T} = \frac{\Delta H_{rev}}{T} \quad -------(7.31)$$

Entropy of fusion:

The heat absorbed, when one mole of a solid melts at its melting point reversibly, is called molar heat of fusion. The entropy change for this process is given by

$$\Delta S_{rev} = \frac{dq_{rev}}{T}$$

$$\Delta S_{fusion} = \frac{\Delta H_{fusion}}{T_f} \quad -------(7.32)$$

where ΔH_{fusion} is Molar heat of fusion. T_f is the melting point.

Entropy of vapourisation:

The heat absorbed, when one mole of liquid is boiled at its boiling point reversibly, is called molar heat of vapourisation. The entropy change is given by

$$\Delta S_v = \frac{\Delta H_v}{T_b} \quad -------(7.33)$$

where ΔH_v is Molar heat of vaporisation. T_b is the boiling point.

Entropy of transition:

The heat change, when one mole of a solid changes reversibly from one allotropic form to another at its transition temperature is called enthalpy of transition. The entropy change is given

$$\Delta S_t = \frac{\Delta H_t}{T_t} \quad -------(7.34)$$

where ΔH_t is the molar heat of transition, T_t is the transition temperature.

Problem: 7.7

Calculate the entropy change during the melting of one mole of ice into water at 0^0 C and 1 atm pressure. Enthalpy of fusion of ice is 6008 J mol^{-1}

Given:

$\Delta H_{fusion} = 6008\ J mol^{-1}$

$T_f = 0^0\ C = 273\ K$

$H_2O(S) \xrightarrow{273\ K} H_2O\ (l)$

$$\Delta S_{fusion} = \frac{\Delta H_{fusion}}{T_f}$$

$$\Delta S_{fusion} = \frac{6008}{273}$$

$\Delta S_{fusion} = 22.007\ J\ K^{-1}\ mole^{-1}$

Evaluate Yourself - 7

Calculate the entropy change when 1 mole of ethanol is evaporated at 351 K. The molar heat of vaporisation of ethanol is 39.84 kJ mol^{-1}

7.11 Gibbs free energy (G)

One of the important applications of the second law of thermodynamics is to predict the spontaneity of a reaction under a specific set of conditions. A reaction that occurs under the given set of conditions without any external driving force is called a spontaneous reaction. Otherwise, it is said to be non-spontaneous. In our day today life, we observe many spontaneous physical and chemical